

For questions 1-3, consider the following context.

Romeo is evaluating the expression $-5(-\sqrt{100} + 25)$. Instead of first performing operations within the parentheses, he multiplied the -5 by both terms in the parentheses, as shown.

$$-5(-\sqrt{100} + 25) = (-5 \cdot -\sqrt{100}) + (-5 \cdot 25)$$

1. Which property of operations did Romeo use?

Distributive property of multiplication over addition.

2. Evaluate both expressions that Romeo set equal to each other using the order of operations.

$$-5(-\sqrt{100} + 25) \rightarrow -5(-10 + 25) \rightarrow -5(15) = \boxed{-75}$$

$$(-5 \cdot -\sqrt{100}) + (-5 \cdot 25) \rightarrow (-5 \cdot -10) + (-125) \rightarrow 50 - 125 = \boxed{-75}$$

3. Explain how the values of the two expressions compare.

They are equal.

For questions 4-5, consider the following context.

Cynthia says that when following the order of operations, the two expressions below are equivalent and the parentheses are not needed.

$$(-2 \cdot \sqrt{64}) + 17 = -2 \cdot \sqrt{64} + 17$$

4. Explain whether Cynthia is correct. Justify your answer mathematically.

$$(-2 \cdot 8) + 17 \rightarrow -16 + 17 = \boxed{1}$$

$$-2 \cdot 8 + 17 \rightarrow -16 + 17 = \boxed{1}$$

Yes, the value of both expressions is 1.

5. Would the parentheses change the value of the expression if they were around $\sqrt{64} + 17$ instead? Justify your answer mathematically.

Yes because parentheses are evaluate before multiplication (order of operations).

$$-2 \cdot (\sqrt{64} + 17) \rightarrow -2 \cdot (8 + 17) \rightarrow -2(25) = \boxed{-50}$$

For questions 6-10, find the value of each expression.

$$6. \quad \left(\frac{1}{7} \cdot 7 - \frac{1}{4}\right) \cdot \sqrt{16}^3$$

$$\rightarrow \left(1 - \frac{1}{4}\right) \cdot \sqrt{16}^3 \rightarrow \frac{3}{4} \cdot 4^3 \rightarrow \frac{3 \cdot 64}{4} \rightarrow \frac{192}{4} = 48$$

$$7. \quad \sqrt{\frac{9^2-9}{2}} - 4.5 + \frac{1}{4}$$

$$\rightarrow \sqrt{\frac{81-9}{2}} - 4.5 + \frac{1}{4} \rightarrow \sqrt{\frac{72}{2}} - 4.5 + \frac{1}{4} \rightarrow \sqrt{36} - 4.5 + \frac{1}{4} \rightarrow 6 - 4.5 + \frac{1}{4} = \boxed{1.75}$$

$$8. \quad \frac{1}{4} \cdot \left(7 - \frac{1}{3}\sqrt{225}\right)^3$$

$$\rightarrow \frac{1}{4} \left(7 - \frac{1}{3} \cdot 15\right)^3 \rightarrow \frac{1}{4} (7 - 5)^3 \rightarrow \frac{1}{4} (2)^3 \rightarrow \frac{1}{4} (8) = \boxed{2}$$

$$9. \quad \sqrt[3]{-125} - 7 \cdot 2^2 + \left(\frac{1}{9}\right)^{-1}$$

$$\rightarrow -5 - 7 \cdot 4 + 9$$

$$\rightarrow -5 - 28 + 9$$

$$= \boxed{-24}$$

$$10. \quad \left(\frac{1}{4} \cdot 16\right)^{-2} - \frac{5}{8} \cdot 24$$

$$\rightarrow (4)^{-2} - \frac{5}{8} \cdot 24$$

$$\rightarrow \left(\frac{1^2}{4^2}\right) - \frac{5}{8} \cdot 24$$

$$\rightarrow \frac{1}{16} - 15$$

$$\rightarrow \frac{1}{16} - \frac{240}{16}$$

$$= \boxed{-\frac{239}{16} \text{ or } -14.9375}$$